MATAR LUDOVIC
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Summary

Research and Development Machine Learning Engineer at French Institute for Research in Computer Science and Automation. Specialized in statistics, probability and **computer science**. Graduated from a **MSc in applied mathematics** and a **MSc in fluid mechanics**.

Work Experiences

Research and Development Machine Learning Engineer at French Institute for Research in Computer Science and Automation (INRIA) - Flowers AI & CogSci Lab. From the 1st of February 2025 to the 31st of July 2026.
Inria Center at the University of Bordeaux, France

- I develop machine learning models to identify sources of interference occurring in simulated multi-core embedded architectures, using automated discovery methodology, that is, curiosity based algorithms.
- Implemented algorithms produce databases with greater diversity measures than other methods. Ongoing development of a semantic behavior space better characterizing interference using BERT.
- Monitoring a 6 months end-of-study internship. From the 1st of April to the 30th of September 2026.

Research Internship at French Alternative Energies and Atomic Energy Commission (CEA) - Laboratory of Artificial Intelligence and Data Science (LIAD, SGLS,DM2S, ISAS, DES). From March to August 2024.
Saclay, Île-de-France, France

- I developed machine learning models to characterize inherent mechanisms of heat exchange until the boiling crisis.
- Works provided research tools for the Laboratory of Modeling and Simulation in Fluid Mechanics (CEA).
- Deep learning methods (computer vision) were used on infrared videos to extract latent space systems of differential equations capturing relevant dynamics by symbolic regression approaches. I used architectures such as CNNs, RNNs combined with autoencoders.
- Python · Deep Learning · Computer vision · Statistics · Fluid mechanics

Research Internship at Renault SAS. From May to September 2023. Guyancourt, Île-de-France, France

- I streamlined the creation of model order reduction techniques to accelerate the evaluation of external aerodynamic performances, making the execution of some computational fluid dynamics analysis optional.
- Obtained models predicting vehicles' drag coefficients with an accuracy higher than 80% using morphing surfaces, numerical simulation results and geometric data.
- Python · Deep Learning · Statistics · Geometry· Fluid mechanics

Tutoring high school students and undergraduates in mathematics since 2023. Paris, Île-de-France, France

Special Skills

- **Languages** : French: native,
  English: C1. **Toefl test on the 8th of April 2026**: reading: 5.5 of 6, listening: 6 of 6, writing: 5 of 6, speaking: 5.5 of 6.
- **Control of development environments** : Solid linux shell + vim. Solid git skills.
- **Python** : Proficient with pytorch, pandas, scikit-learn, numpy, matplotlib.
- **Statistics**: Parametric statistics, hypothesis test, bayesian formulation, time series analysis, non parametric statistics.
- **Machine Learning, Data Science** : Good knowledge of supervised and unsupervised learning, deep Learning, deep reinforcement learning, generative deep learning, natural language processing, LLM skills.

Education

2022 – 2024 Faculté des sciences et ingénierie de Sorbonne Université, Paris
First and second year of master's Degree – Modeling and Simulation in Hydrodynamics (M1 MF2A & M2 MSH, with honors.)

2021 – 2022 Université Paris Cité, Paris
Second year of master's Degree – Random Modeling Data Science (M2 MO, applied mathematics degree.)

2020 – 2021 Université Paris 1 Panthéon – Sorbonne, Paris
First year of master's Degree – Mathematics Applied to Economics & Finance (M1 MAEF, with honors.)

2017 – 2020 Université Paris-Saclay, Orsay
Bachelor's Degree - Mathematics (L3 MINT, with honors.)

Projects I enjoyed

Personal projects: ← **link** **(github)**. My GitHub **Ludovicccccc** shows some selected machine learning and fluid mechanics projects which I've worked on.

- Implementation of Policy gradient, actor critic and double Q learning with a home made grid-world environment.
- Double Q-learning with replay buffer with some open AI gym environments.

Projects:

- Elie Roumet, Ludovic Matar, Raksmy Nop, Geoffrey Daniel, Clément Gauchy, et al.. Boiling phenomena analysis using machine learning. *ICMF 2025 - 12th International Conference on Multiphase flow*, May 2025, Toulouse, France. ⟨cea-05283483⟩
- Ludovic Matar, Clément Moulin-Frier, Eric Jenn, Christophe Marabotto. Application of curiosity driven exploration methods for hardware interference identification. (Expecting a response from conference XXX. Notification on the 20th of April.)

Hobbies

- **Gym and taekwondo**
- **Symphonic Concerts and opera**
- **Piano:** I interpret romantic music.